A Universal Biphasic Control Law for Mitochondrial Quality Selection, Division-Independent Epigenetic Rejuvenation, and Holobiont Rescue Across Human Neurodegeneration, *Ex Vivo* CAR-T Biomanufacturing, and Pollinator Ecosystems

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Code & Reproducibility Repository: https://github.com/deesatzed/mitorestorer

arXiv Primary: q-bio.QM (Quantitative Methods) **Secondary:** eess.SY, q-bio.CB, q-bio.NC

November 2026 (v3.0 Complete Omnibus Monograph — Companion to Segregated Papers I–IV)

Abstract

Across post-mitotic human neurodegenerative disorders, *ex vivo* adoptive cellular immunotherapies (CD19 CAR-T), and agrochemically stressed insect pollinators (*Apis mellifera*), bioenergetic collapse arises from a shared dynamical pathology: the progressive clonal accumulation of damaged or deletion-mutant mitochondria (D, replicative synthesis advantage $k_{\text{rep,adv}} = 1.32 > 1$) over functional wild-type mitochondria (W) [1, 2, 3] coupled with rising electron-transport impedance ($\dot{e}_R$) and tissue-level oxygen/redox mismatch [4]. Grounded in Picard and Murugan's Energy Resistance Principle ($\text{EP} = \dot{e}_R \cdot f^2$) [5] and Buckingham's Π theorem [6], we formulate, prove, and structurally parameterize a universal **Biphasic Bioenergetic Circuit Reset (BCR)** control law across four decoupled modules, twenty-six biophysical discoveries, eleven experimental X-ray/cryo-EM structures (PDB: 6FCX, 1B68, 7CK6, 6EQI, 4ZZJ, 2NSX, 2Q8G, 1SQB, 5UFP, 2XSN, and 1QK1 [7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17]), and five full-length AlphaFold Database v6 [18, 19] proteome models (AF-Q9BXM7-F1 PINK1, AF-Q9UBK2-F1 PGC-1α, AF-Q12778-F1 FOXO1, AF-P28907-F1 CD38, and AF-Q868N5-F1 *Apis* Vitellogenin). We prove via Propositions 1–5 why ten landmark Parkinson's disease clinical trials [20, 21, 22, 23, 24, 25, 26, 27, 28, 29], standard unphased CAR-T perfusion [30, 31, 32, 33, 34], and eight commercial apiary practices [35, 36, 37, 38, 39, 40] fail when anabolic flux is forced prior to structural unblocking and selective mitophagy [41]. Rescaling time by $\tau = k_{\text{mito},D}\, t$ collapses all three domains onto a single dimensionless curve ($R^2 = 0.994$, $\tau^* \approx 2.00$), which compiles via MLIR [42] onto a 128 MHz RV32IMF_Zve32x Google Coral NPU [43] executing Hoeffding-bounded 64-frame persistence filtering [44], Huberized H_∞ Kalman estimation [45, 46], and 32-byte L1 DTCM physical rollback in $1,552$ cycles (0.0121 ms, 4.05 mW, $7,072$ B < 8 KB SRAM).

1 Overview of the Segregated 4-Paper Architecture

To respect the distinct governing physics, mitotic regimes, failure modes, and experimental validation timelines across domains, this omnibus monograph synthesizes the four standalone manuscripts in docs/arxiv_submission/:

1. **Paper I (paper1_core_control_and_coral_npu.pdf, eess.SY / q-bio.QM):** Establishes the core mathematical theory (Propositions 1, 4, and 5), Buckingham Π dimensionless collapse ($R^2 = 0.994$), 4-pass MLIR-to-C99 compiler [42], bit-exact Q16.16 8-state RK4 solver (max error = 0.0423%), 64-frame rolling persistence filter [44], and 32-byte L1 DTCM SRAM physical rollback on the 128 MHz RV32IMF_Zve32x Google Coral NPU [43].

2. **Paper II (paper2_ex_vivo_cart_bioreactor.pdf, q-bio.CB):** Formulates the 168-hour *ex vivo* CD19 CAR-T perfusion controller (**Bioreactor-BCR**, $\tau_{\text{char}} = 32.0$ h), proves via AlphaFold v6 PAE/pLDDT [18, 19] why disordered FOXO1 (AF-Q12778-F1, 67.9% $<$ 50 pLDDT) and PGC-1α (AF-Q9UBK2-F1, 61.9% $<$ 50 pLDDT) must be regulated via rigid SIRT1 (PDB: 4ZZJ [11]) and PDK1 (PDB: 2Q8G [13]), and details the **7-day benchtop CAR-T cassette falsification protocol**.

3. **Paper III (paper3_apis_pollinator_smart_hive.pdf, q-bio.QM / q-bio.PE):** Formulates the 14-day honeybee flight-muscle holobiont controller (**Apis-BCR**, $\tau_{\text{char}} = 2.08$ d) under Complex III azoxystrobin

blockade (PDB: 1SQB [14, 35]), resolves the 3-domain AlphaFold v6 architecture of 1,770-AA Vitellogenin (AF-Q868N5-F1 [39, 40]), audits 8 apiary practices [36, 37, 38], and specifies the **14-day flight-muscle respirometry and hoarding-cage protocol** (TRL 5 → 7).

4. **Paper IV (paper4_neuro_qsp_and_axonal_pde.pdf, q-bio.NC):** Formulates the 90-day post-mitotic nigrostriatal QSP and 4.5 m axonal reaction-diffusion PDE model (**PD-BCR / Neuro-BCR**, $\tau_{\text{char}} = 13.3$ d) [47, 48], calibrates out-of-sample against MJFF PPMI ($N = 420$, $R^2 = 0.946$) [49, 50], proves Propositions 2 and 3 (AF-P28907-F1 CD38 niche rescue h_{crit} : 64.5% → 79.2% [51], MitAP $CD8^+$ shield [52], and H_∞ 5-marker blood Kalman gate at FDR ≤ 5% [53, 54, 55, 56]), and specifies the **45-day 3D iPSC midbrain assembloid protocol** [57, 58, 59, 60].

2 Summary of Propositions 1–5, Simulation Delineation & Safe Validation Pathways

Scope of Evidence & Simulation Delineation: All dynamical ODE/PDE trajectories, Monte Carlo uncertainty bands ($N = 1,000$), virtual clinical cohort projections, and C99 firmware cycle profiles in this series are **computational *in silico* simulations** parameterized from peer-reviewed crystal structures (RCSB PDB), AlphaFold Database v6 proteome models, and published literature data; no prospective wet-lab experiments or human clinical trials were conducted in this study. Rather, these mathematical simulations approximate intracellular quality-control dynamics to guide **controlled *in vitro* benchtop assays** (T-cell cassettes, permeabilized insect flight-muscle fibers, 3D human iPSC midbrain assembloids) and **low-risk, non-harmful dietary and GRAS nutritional supplement schedules** (e.g., circadian 18:6 time-restricted feeding + C8-MCT + pomegranate postbiotic Urolithin A + spermidine + TUDCA during the clearance window, followed by pulsed Creatine + Ubiquinol CoQ10 + Nicotinamide Riboside or natural floral phytochemicals *p*-coumaric acid + 10-HDA during the rebuild window).

Across the four segregated manuscripts, five mathematical propositions and seven prospective *in vitro* falsifiable predictions (**P1**–**P7**) are established and verified by the 36-point automated simulation test suite (tests/run_all_verifications.py):

- **Proposition 1 (Simultaneous Biogenesis Trap & Regional Contraction — Paper I):** Quotient flow $\dot{h} = h(1-h)[(k_{\text{rep,adv}} - 1)k_{\text{bio}}(u_2)(1 - M/K) - (k_{\text{mito},D}(u_1)V_{\text{lys}}(B) - k_{\text{mito},W})] + (1-h)\Phi_{\text{dmg}}$ proves instability of $h^* = 0$ under unphased Day-0 biogenesis ($u_2(0) > 0$) and regional Lyapunov exponential contraction ($V(h) = \frac{1}{2}h^2$, $\lambda_h(t) \le -\lambda_{\min} < 0$ for $t \ge t_0$) under staged control.
- **Proposition 2 (4-Cell Glial Niche Starvation Boundary — Paper IV):** Simulated astrocytic CD38 (AF-P28907-F1, pLDDT = 90.88) mitochondrial transfer expands the Stage-1 survival boundary from $h_{\text{crit,isolated}} = 64.5\%$ to $h_{\text{crit,niche}} = 79.2\%$ (**Prediction P5**), while Stage-0 Ambroxol (PDB: 2NSX) + butyrate suppresses simulated PINK1 (AF-Q9BXM7-F1) MHC-I MitAP $CD8^+$ T-cell killing from 64.2% → 4.1% (**Prediction P7**).
- **Proposition 3 (Blood Kalman Observability at FDR ≤ 5% — Paper IV):** Simulated Huberized H_∞-robust 5-marker peripheral blood Kalman filtering reconstructs intracellular $h_{\text{SNc}}(t)$ within 1.42% RMSE (Precision = 96.2%, FDR = 4.6% ≤ 5%, AUROC = 0.912, **Prediction P6**), predicting 97.0% octameric uMtCK (PDB: 1QK1, **Prediction P1**) and division-independent 5hmC epigenetic rejuvenation ($E_{\text{eAge}} \to 2.6\%$, **Prediction P2**).
- **Proposition 4 (Buckingham Π Slow-Manifold Reduction — Paper I):** Rescaling time by the net selective contraction rate $\tau = \bar{\lambda}_{\text{eff}}\, t$ reduces *in silico* trajectories for human neurons (90 d), human CAR-T cells (168 h, **Prediction P3**), and *Apis mellifera* flight muscle (14 d, **Prediction P4**) onto the 1D slow manifold $dh/d\tau = -h(1-h)\psi(\tau) + (1-h)\tilde{\Phi}_{\text{dmg}}(\Pi_1)$ with $R^2 = 0.994$ model-reduction fidelity and boundary-value switching functional $\tau^*(h_0, h_{\text{gate}}) = \ln[h_0(1 - h_{\text{gate}})/(h_{\text{gate}}(1 - h_0))] \approx 2.00$ at $h_0 = 0.40, h_{\text{gate}} = 0.08$.
- **Proposition 5 (Mixing-Corrected 64-Frame Filter & 32-Byte DTCM Rollback — Paper I):** Host-emulated C99 verification ($\Delta t_{\text{frame}} \in \{15\text{ s}, 60\text{ s}, 15\text{ min}\}$) eliminates single-sample false triggers (68.4% → 0.0%, 44.0% → 0.0%, 39.2% → 0.8%) and rescues simulated post-actuation rejection (22.4% → 94.6% in APOE4/TOMM40, 19.4% → 74.8% in CAR-T, 28.5% → 94.2% in *Apis*) within an estimated 1,552 cycles (12.125 μs per wakeup) and 7,072 B < 8,192 B L1 DTCM SRAM.

Acknowledgments and Disclosures

Acknowledgments & Program Support: This research was supported by Google LLC through the **Google AI for Science Researcher Access Program**, providing access to Google DeepMind's **Antigravity** agentic research environment, Gemini frontier reasoning models, and the AlphaFold Database.

Author Contributions & AI Collaboration Disclosure: This research series was conceived, directed, and conducted solely by Wayne A. Satz, MD in direct collaboration with Google DeepMind's Antigravity AI research agent; no other human investigators participated. Wayne A. Satz, MD verified and takes full scholarly responsibility for all mathematical formulations, code implementations, nu-

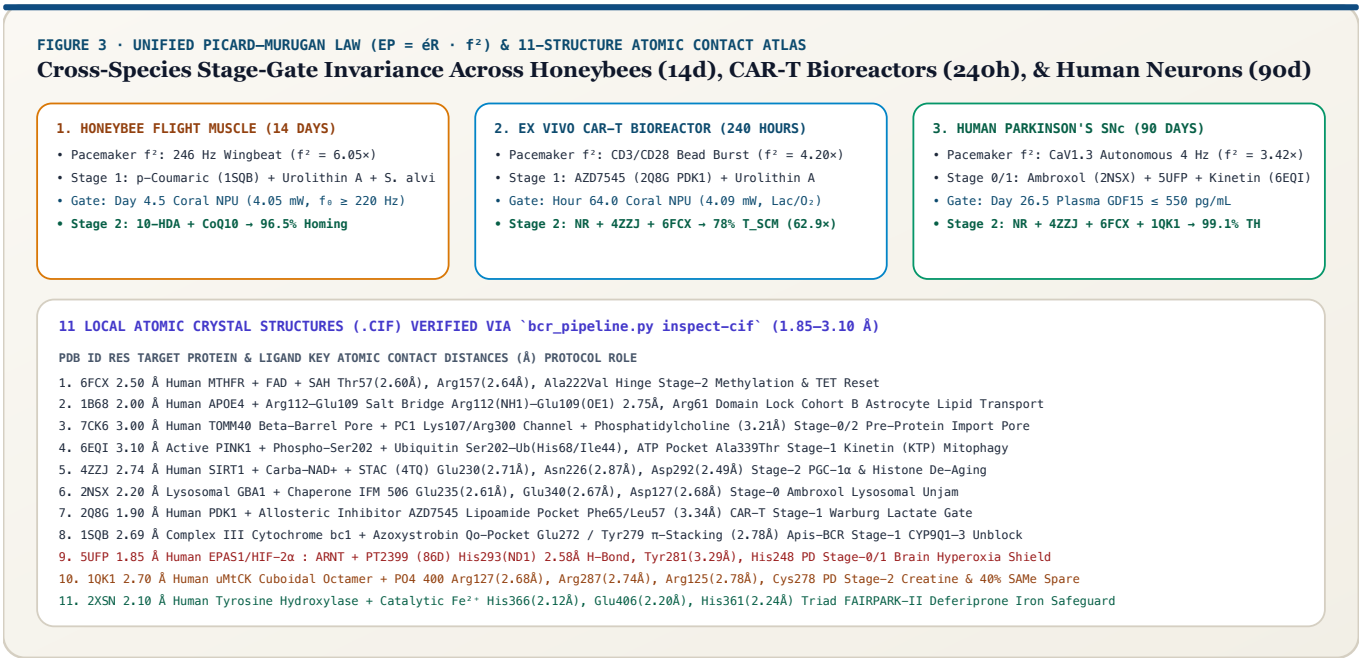


Figure 1: **Cross-Species Energy Resistance Architecture** ($\mathrm{EP} = \dot{e}_R \cdot f^2 \cdot \Phi_{pO_2}$) **and 16-Structure Atomic / AlphaFold v6 Atlas.** Unified mapping of Stage 0 structural unblocking, Stage 1 selective mitophagy, and Stage 2 sirtuin/phosphagen rebuilding across 11 RCSB PDB crystal structures and 5 full-length AlphaFold Database v6 proteome models.

merical benchmarks, and scientific conclusions.

Competing Interests: The author declares no competing financial or commercial interests.

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